



UNIVERSITÀ
di **VERONA**

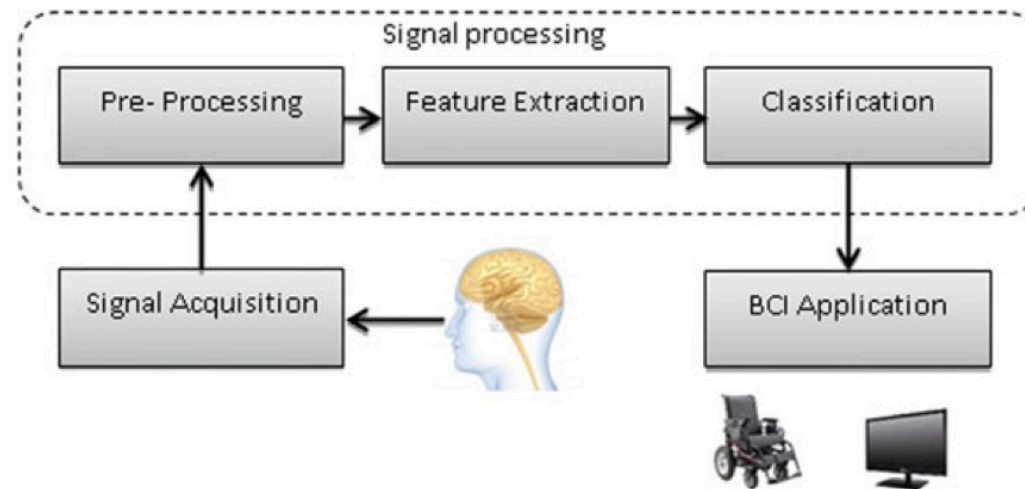
PHD School
in **NATURAL SCIENCES
AND ENGINEERING**

Brain Computer Interfaces

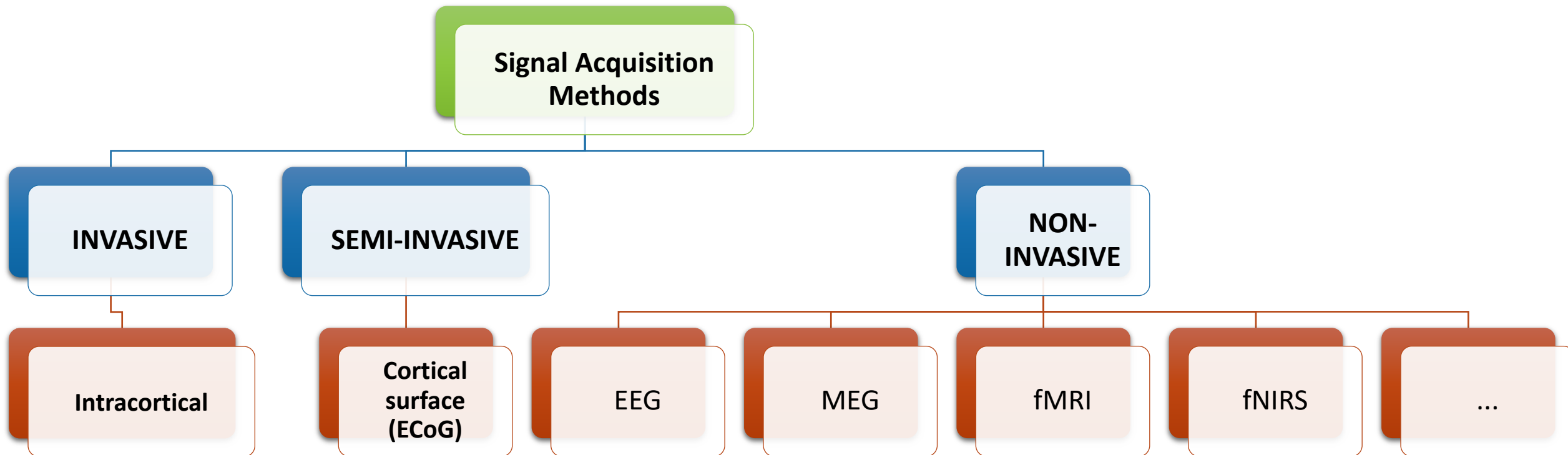
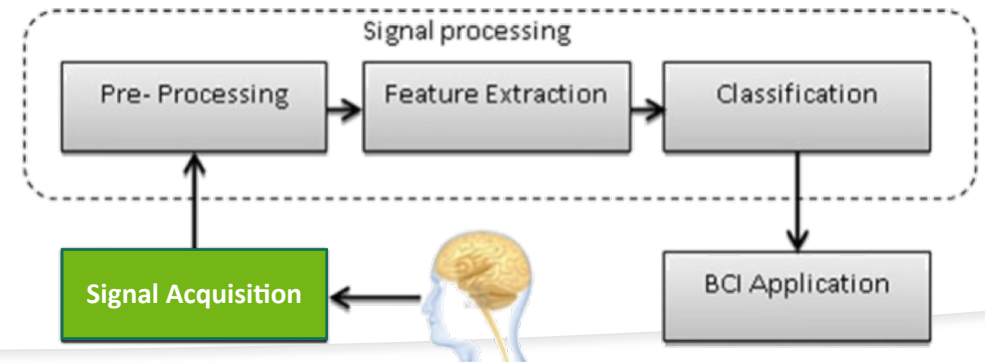
Lecture 2: Design and implementation of a BCI

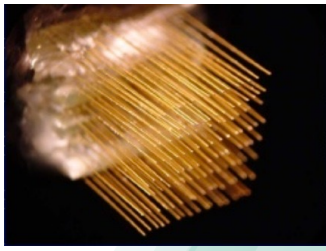
Design and implementation of a BCI

- **Signal Acquisition**
- **Pre-processing**
- **Feature extraction**
 - It depends on the process of interest
(e.g. oscillation, ERP peak, complex phenomenon)
- **Classification**
- **BCI application**



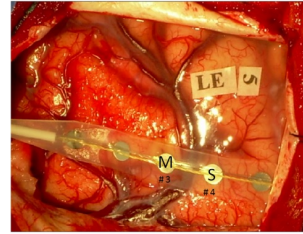
Signal acquisition





INVASIVE

- It requires the implant of microelectrode **directly into the subject's brain** through surgery
- It produces signals with high SNR
- Invasive BCI devices are used to provide functionality to paralyzed people



SEMI-INVASIVE

- A strip of electrodes is placed **on the surface of the brain** via surgery in particular in the motor cortex or premotor of subjects paralyzed
- It allows the recording of the signal on a greater surface area of the cortex allowing the distinction of different cortical activation pattern
- It has less risk of scar tissue formation when compared to invasive BCI



NON-INVASIVE

- The brain signals are recorded by means of non-invasive techniques (e.g. EEG, fMRI, fNIRS)
- One of most commonly used is based on real-time data obtained from the EEG

Invasive vs. Non-invasive

PROS

- High SNR
- Reduction of training times
- Greater number of degrees of freedom to control
- Best performance, as regards the signal classification accuracy and the speed of information transmission

CONS

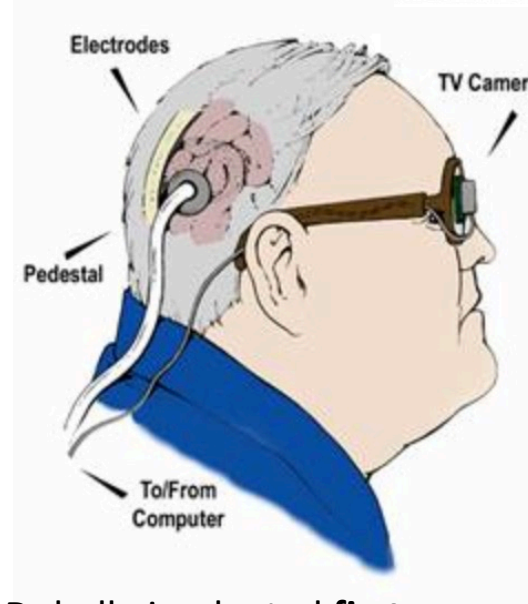
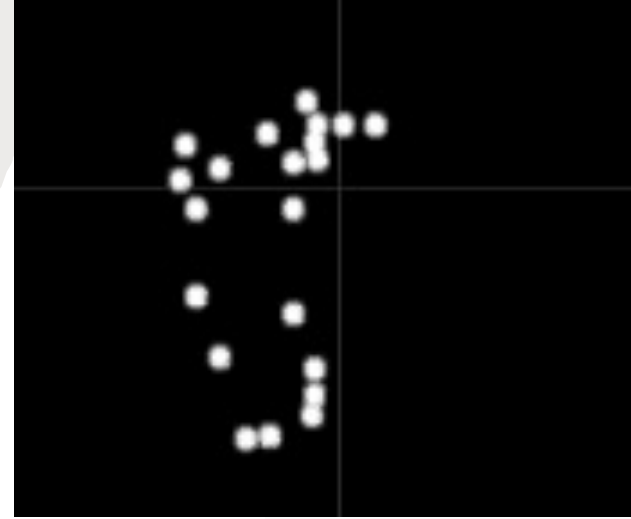
- Requires the complex procedure of implantation of the electrodes, (only for subjects with severe disabilities)
- Expensive systems

Invasive BCI - visual prosthesis

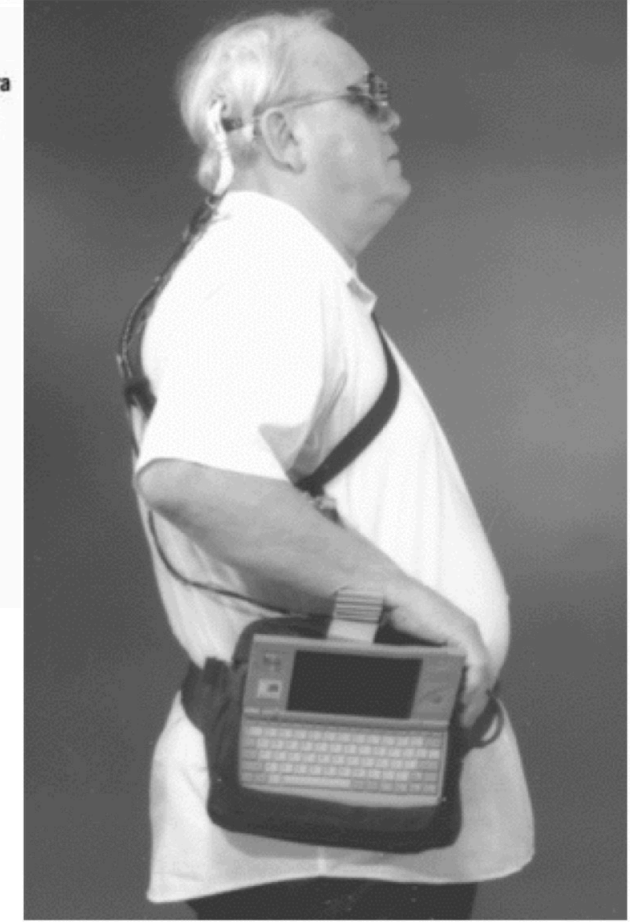
Dr. W.H. Dobelle (1941–2004) was a biomedical researcher who developed experimental technologies that restored limited sight to blind patients

The Dobelle implant

- Was developed for blind people, with the objective of transmitting real-time visual information from a camera to the brain
- Uses a digital video camera mounted on a pair of glasses as an input source for imaging
- Sends this signal to a computer which processes this information and outputs this to the visual cortex
- Allows the subject to see a pattern of phosphenes, or flashes of light



Dobelle implanted **first prototype** into “Jerry”, a man blinded in adulthood, in 1978.



EEG-based BCIs

Commercialized EEG devices for BCI applications

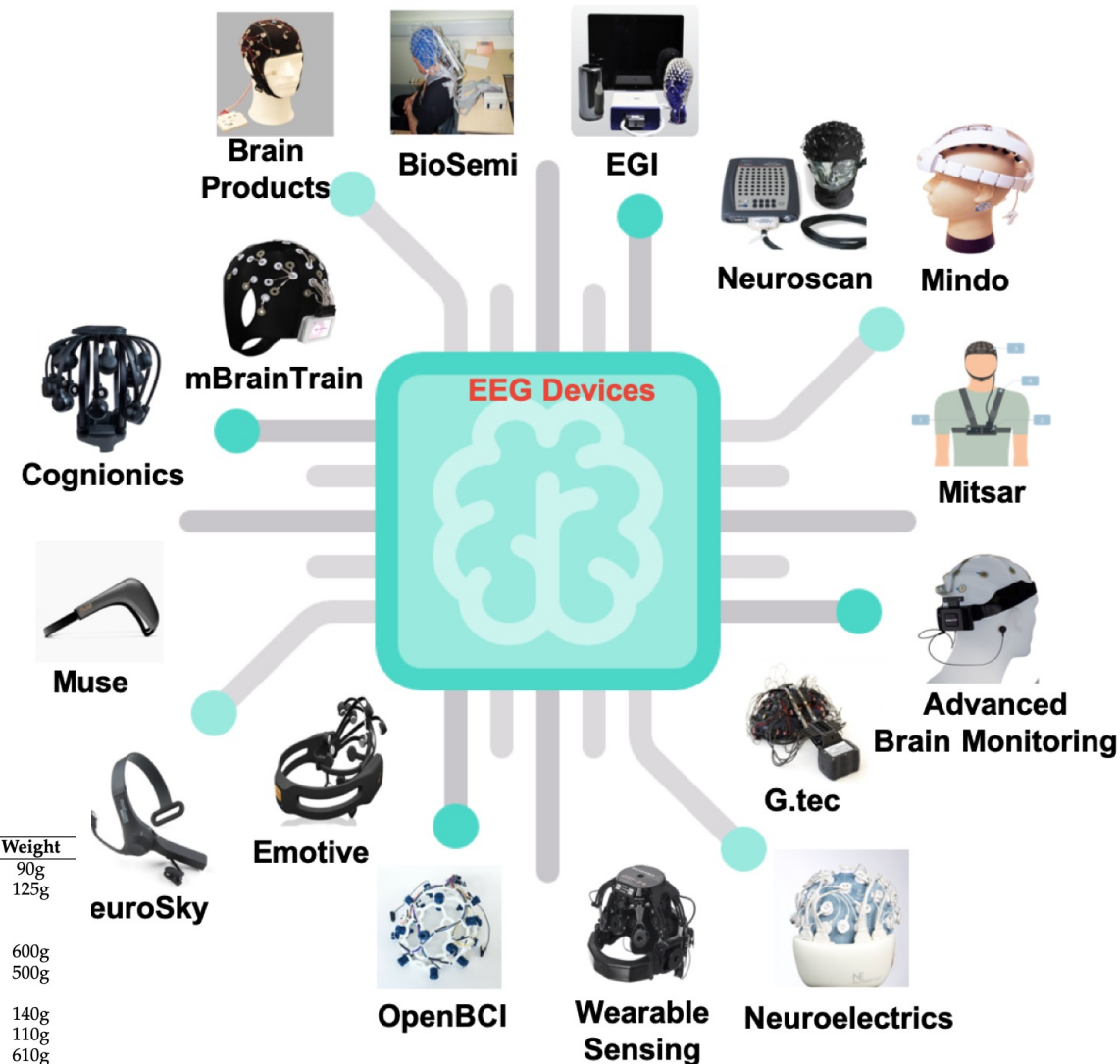


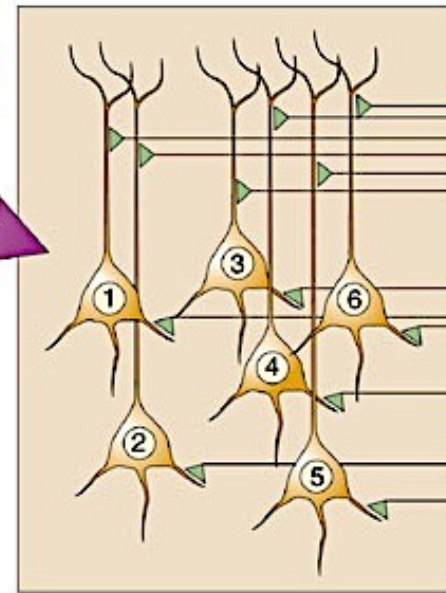
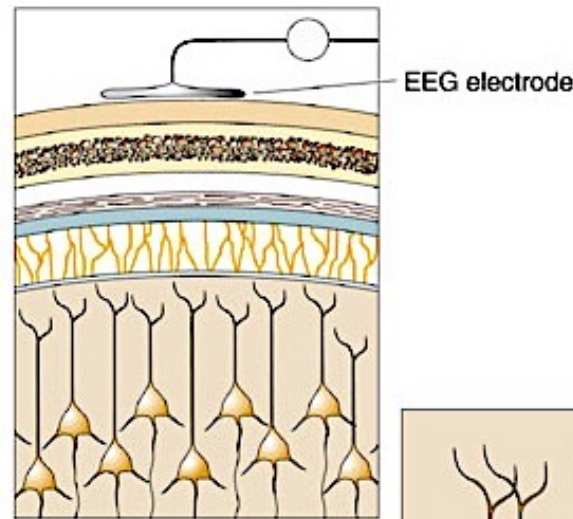
TABLE 1
An Overview of EEG Devices

Brand	Product	Wearable	Sensors type	Channels No.	Locations	Sampling rate	Transmission	Weight
NeuroSky	MindWave	Yes	Dry	1	F	500 Hz	Bluetooth	90g
Emotiv	EPOC(+)	Yes	Dry	5-14	F, C, T, P, O	500 Hz	Bluetooth	125g
Muse	Muse 2	Yes	Dry	4-7	F, T		Bluetooth	
OpenBCI	EEG Electrode Cap Kit	Yes	Wet	8- 21	F, C, T, P, O		Cable	
Wearable Sensing	DSI 24; NeuSenW	Yes	Wet; Dry	7-21	F, C, T, P, O		Bluetooth	600g
ANT Neuro	eego mylab / eego sports	Yes	Dry	32 - 256	F, C, T, P, O	Up to 16 kHz	Wi-Fi	500g
Neuroelectrics	STARSTIM; ENOBIO	Yes	Dry	8-32	F, C, T, P, O	125-500 Hz	Wi-Fi; USB	
G.tec	g.NAUTILUS series	Yes	Dry	8-64	F, C, T, P, O	500 Hz	Wireless	140g
Advanced Brain Monitoring	B-Alert	Yes	Dry	10-24	F, C, T, P, O	256Hz	Bluetooth	110g
Cognionics	Quick	Yes	Dry	8-30; (64-128)	F, C, T, P, O	250/500/1k/2k Hz	Bluetooth	610g
mBrainTrain	Smarting	Yes	Wet	24	F, C, T, P, O	250-500 Hz	Bluetooth	60g
Brain Products	LiveAmp	Yes	Dry	8-64	F, C, T, P, O	250/500/1k Hz	Wireless	30g
Brain Products	AntiCHapmp	Yes	Dry	32-160	F, C, T, P, O	10k Hz	Wireless	1.1kg
BioSemi	ActiveTwo	No	Wet (Gel)	280	F, C, T, P, O	2k/4k/8k/16k Hz	Cable	1.1kg
EGI	GES 400	No	dry	32-256	F, C, T, P, O	8k Hz	Cable	
Compumedics Neuroscan	Quick-Cap + Grael 4k	No	Wet	32-256	F, C, T, P, O		Cable	
Mitsar	Smart BCI EEG Headset	Yes	Wet	24-32	F, C, T, P, O	2k Hz	Bluetooth	50g
Mindo	Mindo series	Yes	Dry	4-64	F, C, T, P, O		Wireless	

Abbreviation: Frontal (F), Central (C), Temporal (T), Partial (P), and Occipital (O)

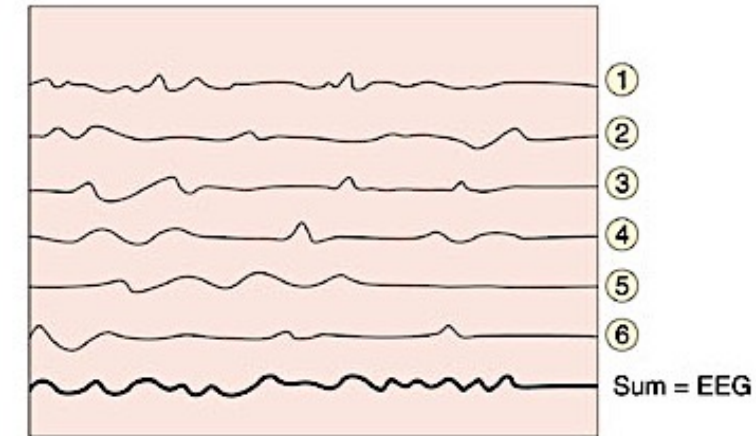
Origin of EEG signal

- EEG requires synchronous activity across groups of cells
- Synchronized neural activity produces larger signals
- Deep sources (subcortical areas) are much more difficult to detect than currents near the skull



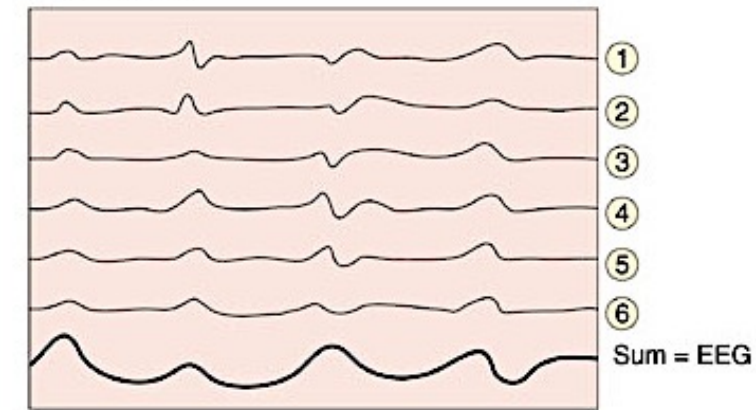
(a)

Irregular



(b)

Synchronized



(c)

The EEG signal changes

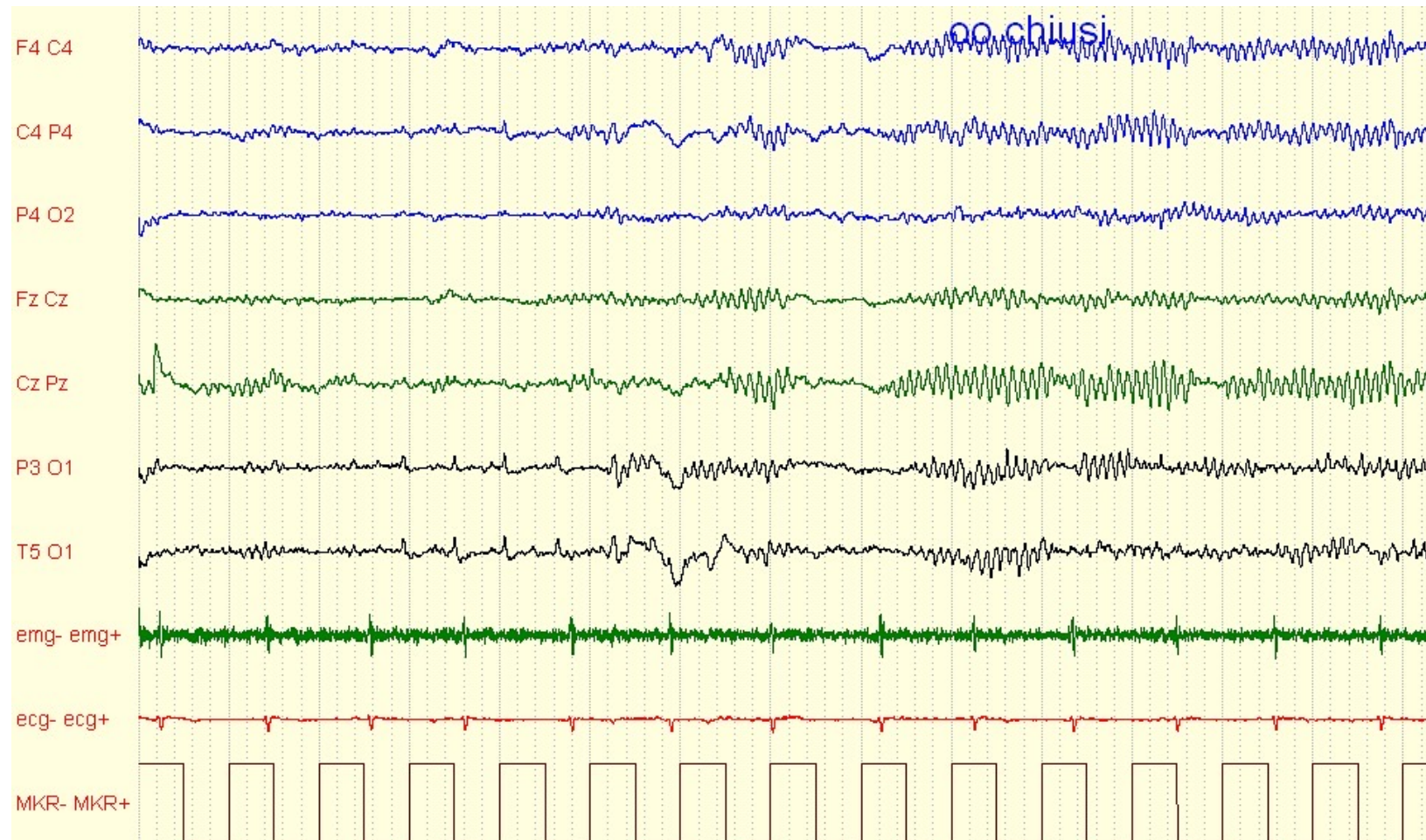
It is an expression of potential changes of the brain cortex and varies from point to point in the cortical surface.

Variations due to:

- Open-closed eyes
- Seated or lying
- Non-thinking or mental calculation
- Rest or motion

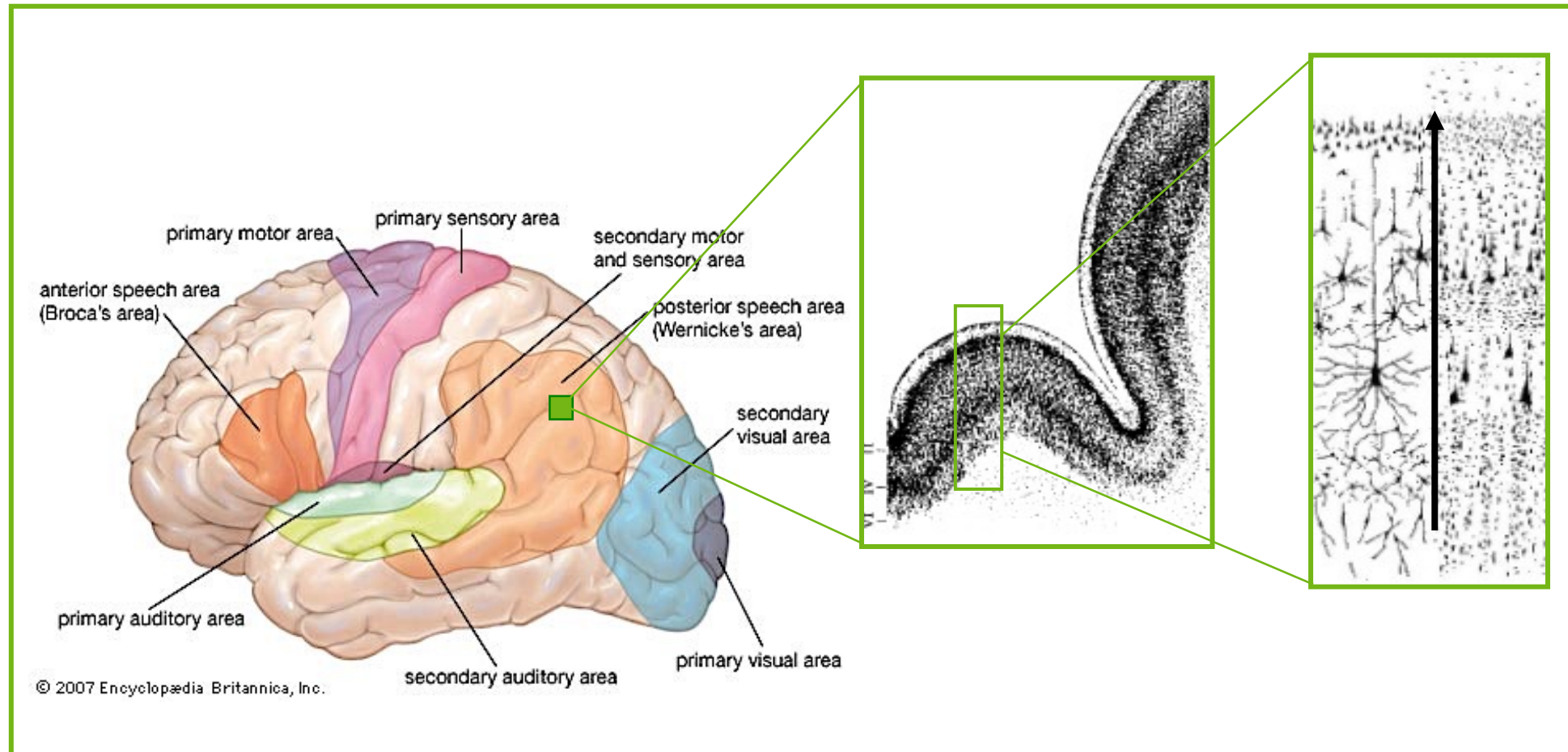
Inter-individual and intra-individual variability

- Peak variable 8-13 Hz
- Age



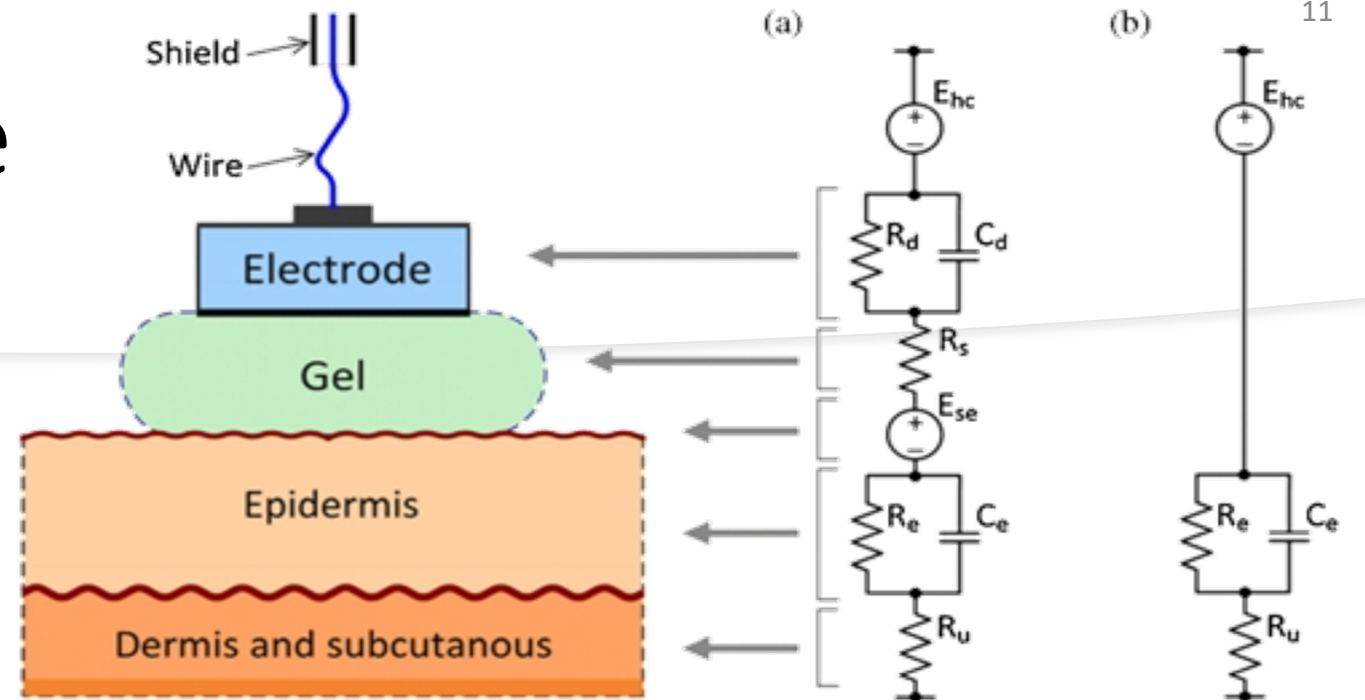
Physiological sources of EEG activity

- **EEG is the recording of electrical activity along the scalp**
- EEG signals derive primarily from cortical current sources
- EEG measures voltage fluctuations resulting from ionic current flows within the neurons of the brain



Electrode impedance

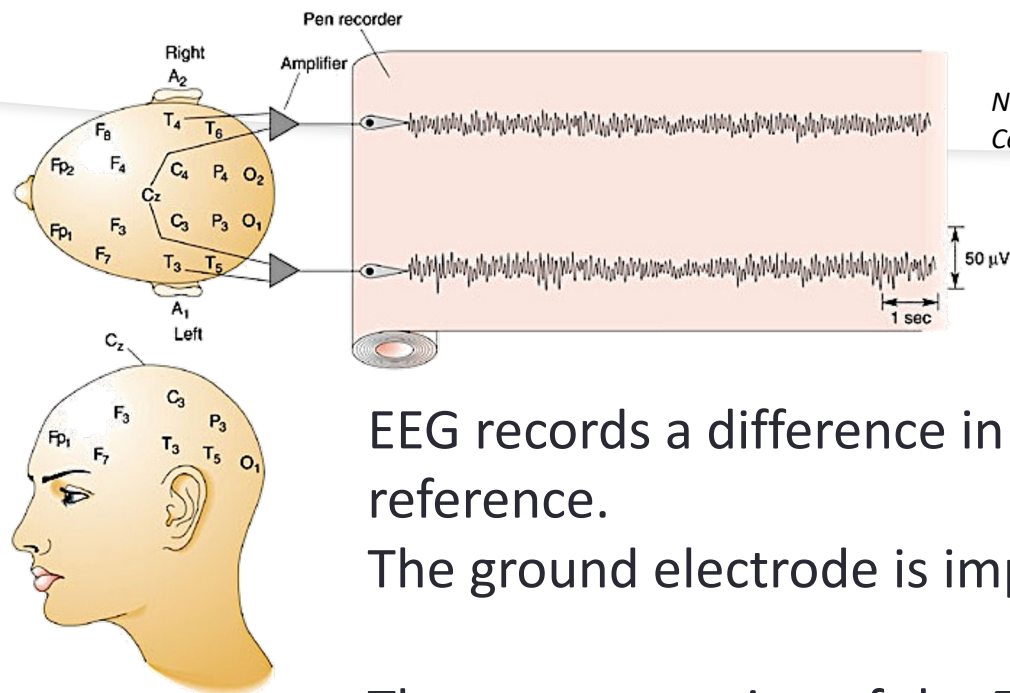
The quality of the electrode contact is quantified by its **electrical impedance**



Meziane et al 2013 Physiol. Meas.

- Dry and/or old skin creates a high impedance, which makes it difficult to acquire good recordings
- Clinicians rub the skin with a mild abrasive to remove the thin layer of dead skin to enable better ion flow between the tissue and the electrolyte on the electrode
- Problems occur when the electrolyte dries over the course of several hours
- The required procedure is to abrade the skin to achieve a scalp-electrode impedance of less than 5 k Ω

EEG recording mode



Neuroscience: Exploring the Brain, 3rd Ed, Bear, Connors, Paradiso 2007

EEG records a difference in potential at the scalp using a set of active electrodes and a reference.

The ground electrode is important to eliminate noise from the amplifier circuit.

The representation of the EEG channels is referred to as a **MONTAGE**

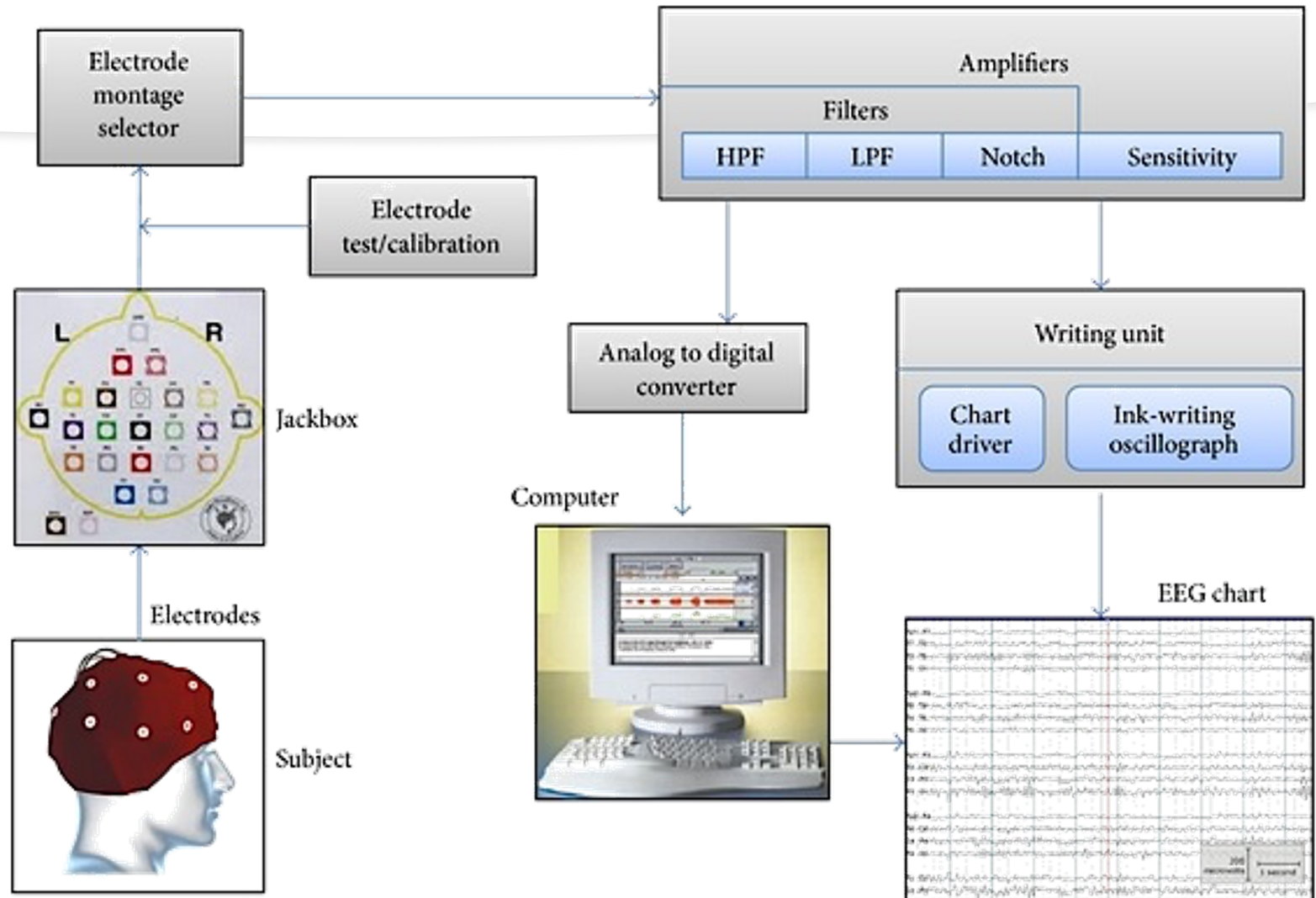
- **MONOPOLAR:** potential difference between an “active” electrode and designed reference
- **BIPOLAR:** potential difference between two “active” electrodes

Electrodes measure voltage-differences at the scalp in μV range

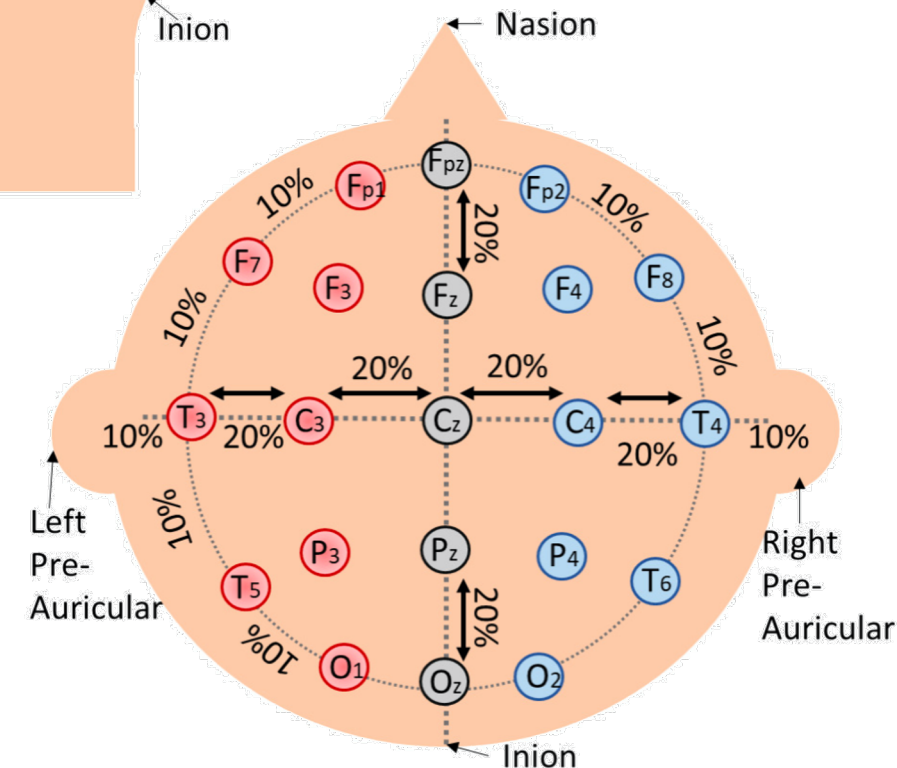
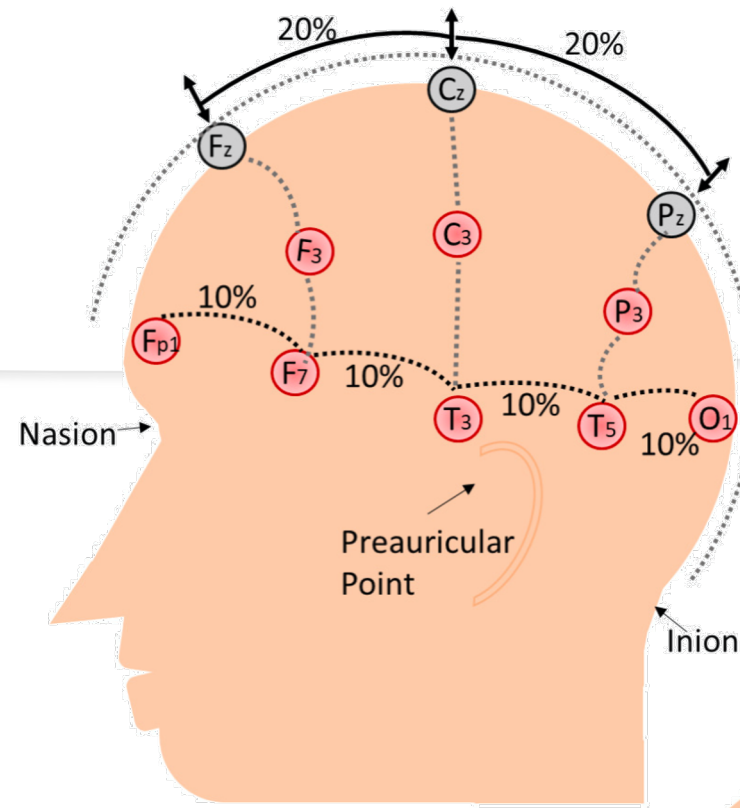
Voltage-traces are recorded with ms resolution

Digital EEG system

- Filtering
- Amplification
- Sampling and analogic to digital converter
- Graphical interface
- Visualization of EEG signal



Standard Montage



International 10-20 system

The "10" and "20" refer to the fact that the actual distances between adjacent electrodes are either 10% or 20% of the total front-back or right-left distance of the skull.

- Reproducibility
- Between subjects comparison

Standard Montage

Symmetry

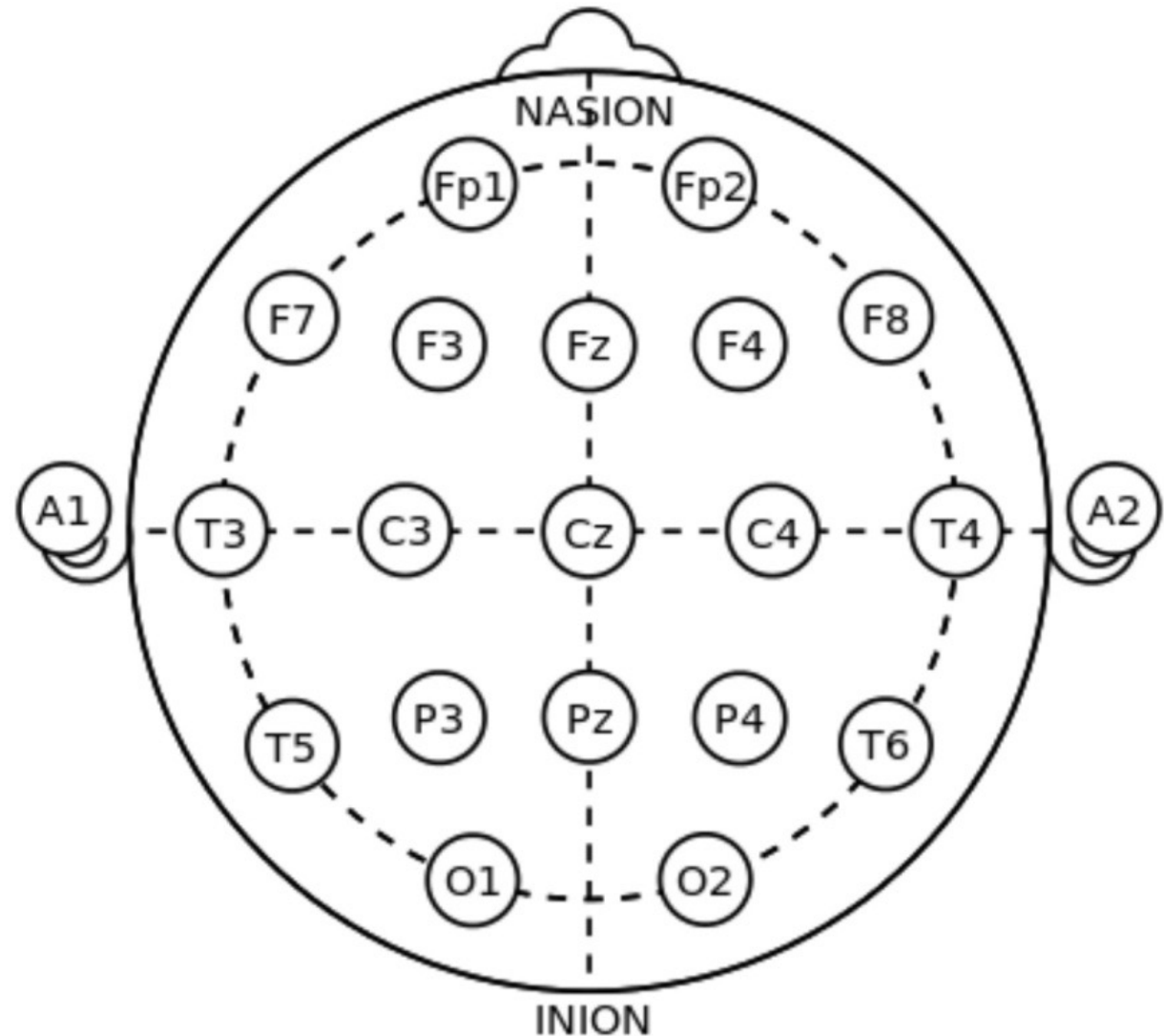
- Antero-posterior
- Between hemispheres

Location

- Each site has a letter to identify the lobe and a number to identify the hemisphere location
- The letters F, T, C, P and O stand for frontal, temporal, central, parietal, and occipital lobes, respectively

Label

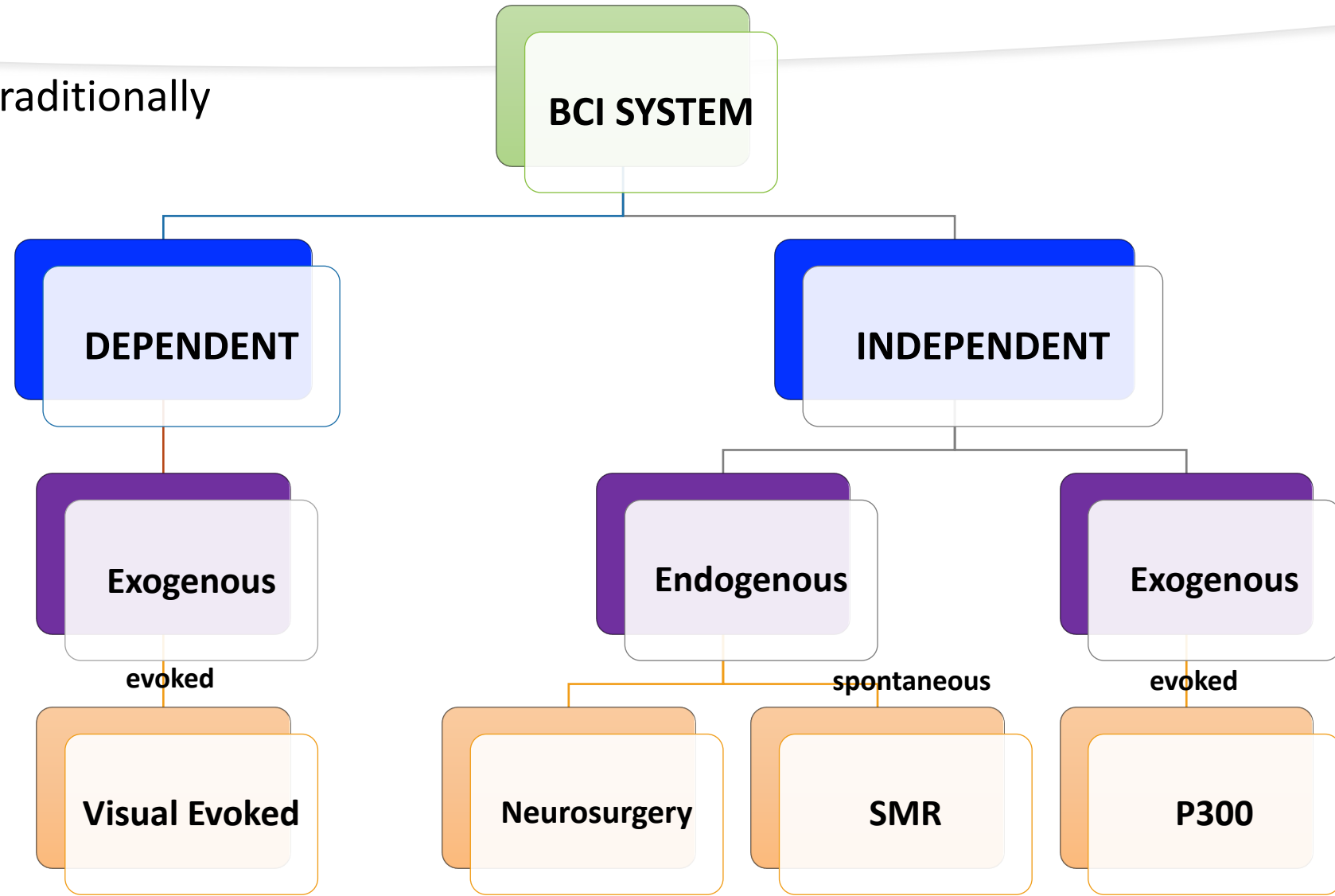
- A “z” refers to an electrode placed on the midline.
- Even numbers – right hemisphere
- Odd numbers – left hemisphere



Types of BCI

The development of the BCI is traditionally divided into several categories:

- independent, dependent
- exogenous, endogenous
- evoked, spontaneous



DEPENDENT

- Does not use the brain's normal output pathways to carry the message, but **activity in these pathways is needed to generate the brain activity** (e.g. EEG) that does carry it
- Replaces a way interrupted or deteriorated with another pre-existing, which is used to generate the brain activity

- Does not depend on the brain's normal output pathways (peripheral nerves and muscles), and **activity in these pathways is not needed to generate the brain activity**
- Is more useful than the BCI dependent for people with severe neuromuscular disabilities, which cannot communicate with the normal ways (including the extraocular muscle control)
- Is of greater theoretical interest because it bypass the peripheral activities

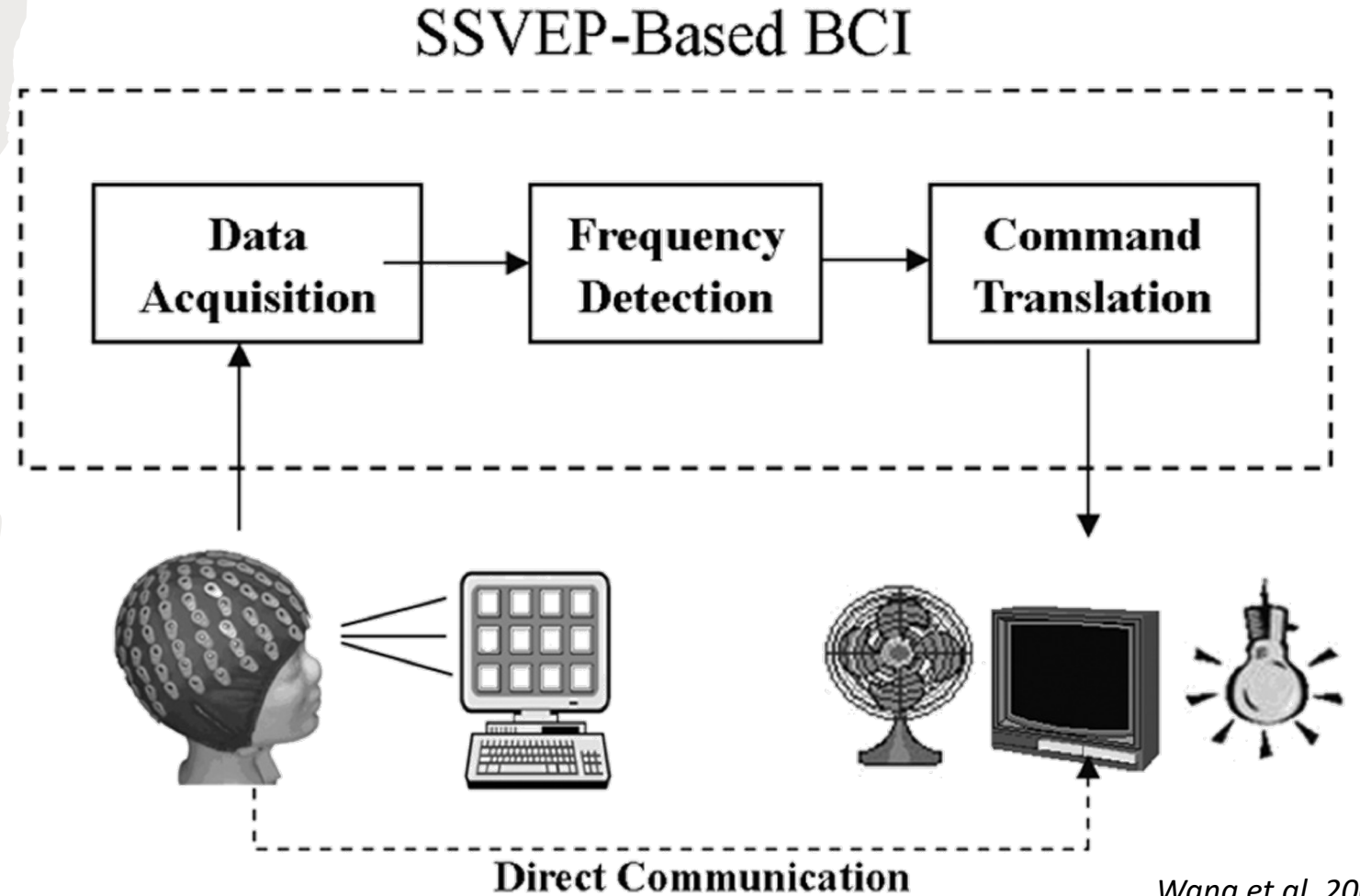
INDEPENDENT

Example (1 of 2)

BCI dependent

- **Matrix of letters** that flash one at a time (rows or columns)
- The user selects a specific letter by looking directly at it so that the **VEP** recorded from the scalp over visual cortex when that letter flashes is much larger than the VEPs produced when other letters flash
- The brain's output channel is EEG, but the generation of the EEG signal depends on **gaze direction**, and therefore on extraocular muscles and the cranial nerves that activate them

Sutter, 1992

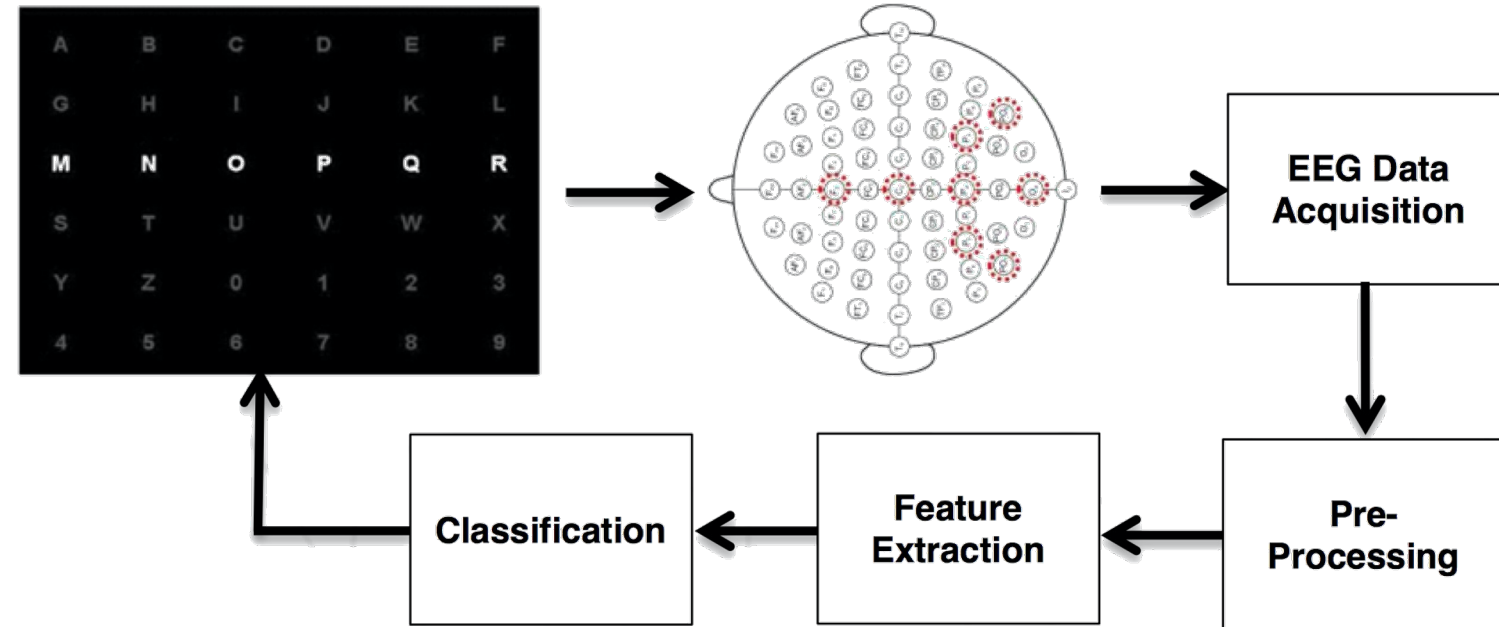


Wang et al. 2006

Example (2 of 2)

BCI independent

- **Matrix of letters** that flash one at a time (rows or columns)
- The user is instructed to select a character when that letter flashes by focusing on it evoking **P300 response** in EEG
- The brain's output channel is EEG, and the generation of the EEG signal depends mainly on the **user's intent**, not on the precise orientation of the eyes
- The normal output pathways of peripheral nerves and muscles do not have an essential role in the operation of an independent BCI



Setare Amiri et al. 2013

EXOGENOUS

- Depends on neuronal activity evoked by external stimuli (e.g. VEP or AEP)
- Does not require intensive training, their control signals are easy to evoke
- Its control signals can also be detected with a few EEG channels

Training

- *neurofeedback*
- Its duration ranges from subject to subject, type of experiment and experimental environment

- The subject voluntarily produces the required signal
- Is not based on an external stimulus, but it is mainly based on brain rhythms and other potentials (**motor imagery, computational tasks**)
- The subject learns the ability to produce specific patterns that are then decoded by the system

ENDOGENOUS

EVOKED

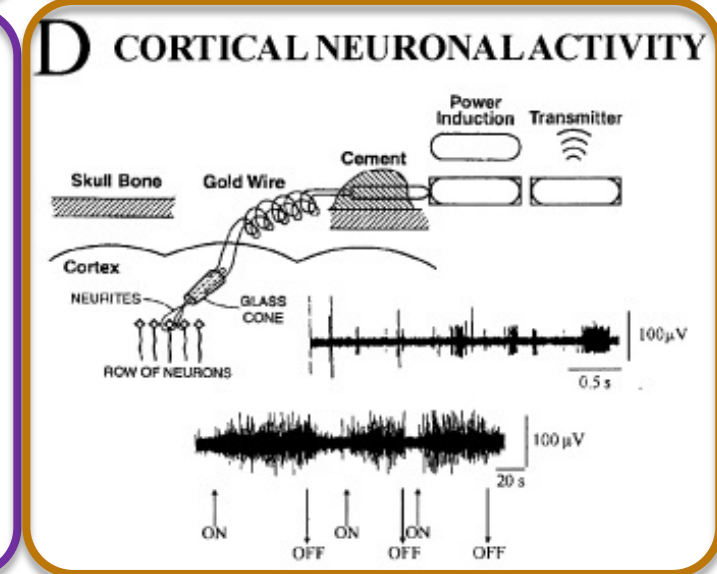
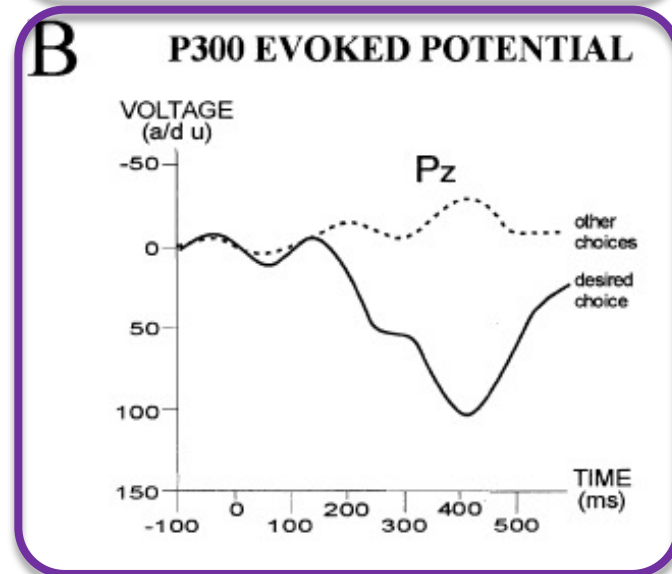
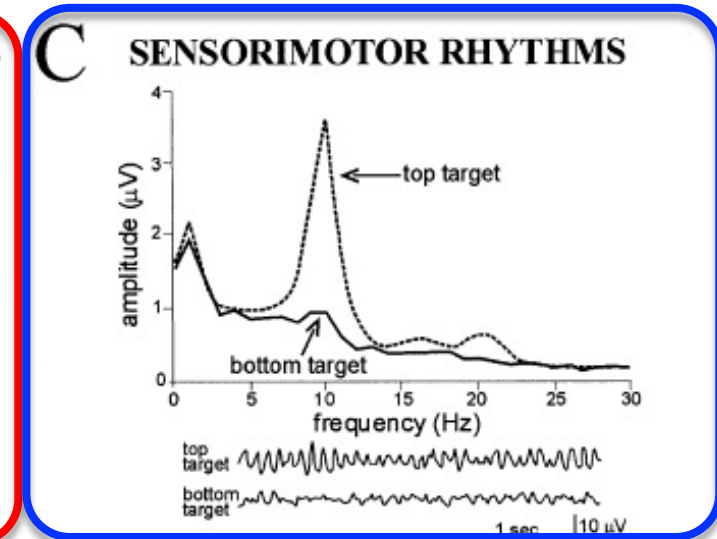
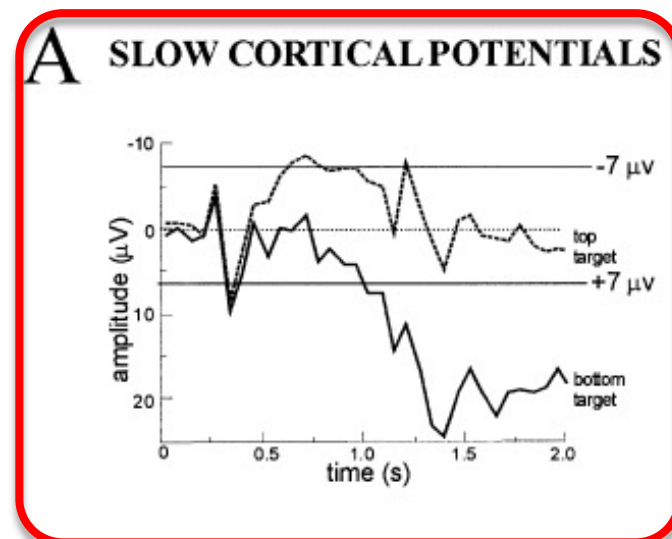
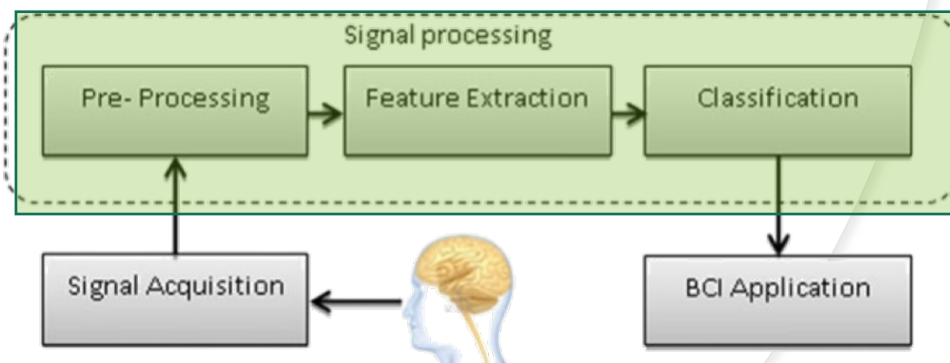
- It depends on the type of evoked potential, which reflects the immediate automatic response of the brain to some external stimuli
- The P300 and the steady-state visual potential (SSVEP) are the most commonly explored in BCI research
- The need for external stimulation can limit the applicability of the evoked potentials to a limited range of tasks

- The spontaneous BCI systems allow the user to freely do cognitive processes, as it eliminates the need of external stimulation
- This allows the user to interact with the system in a more natural way through the intent
- This method is particularly useful when controlling robotic devices
- Self regulated signals: ERD/ ERS; slow cortical potentials (SCP)

SPONTANEOUS

CONTROL SIGNALS

EEG-based BCI

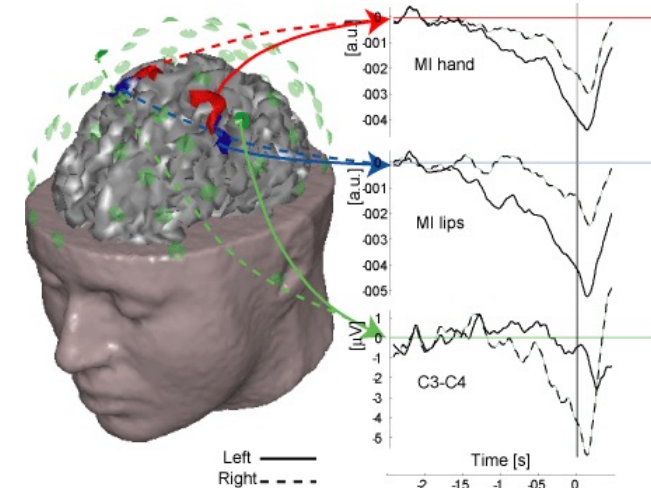


SCP

The **Thought-Translation-Device (TTD)** is an EEG-based BCI developed to re-establish communication in severely paralyzed subjects

- First successful device: end 1990's
- TTD relies on the **self-regulation** of SCPs
- The user can learn to self-control their SCPs when they are provided with **visual or auditory feedback** of their brain potentials and when potential changes in the desired direction are positively reinforced
- In TTD the vertical position of a feedback cursor reflects the amplitude of an SCP shift
- After a patient has achieved reliable control over his/her SCP shifts, the responses can be used to select items presented on a computer screen
- **Intensive training** is necessary to gain control over the SCP waves

Patient using TTD writes a letter



EVOKED POTENTIAL (EP)

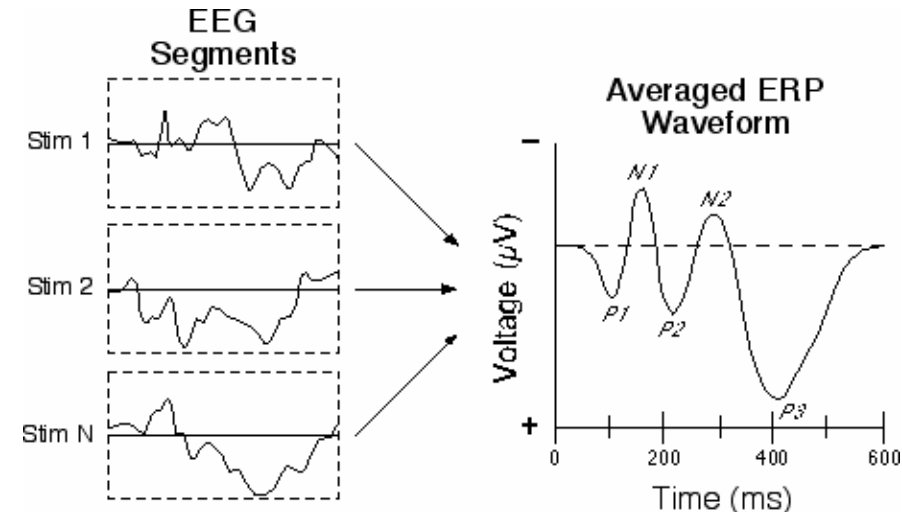
An **evoked potential** is an **electrical potential recorded from the nervous system in response to stimulation of specific sensory nerve pathways.**

Evoked potential amplitudes ranging from less than a μV to several μV .

$$y(t) = u(t) + v(t)$$

Diagram illustrating the components of the evoked potential signal:

- $u(t)$ (blue circle) is labeled **EP** (Evoked Potential).
- $v(t)$ (red circle) is labeled **noise (EEG)**.



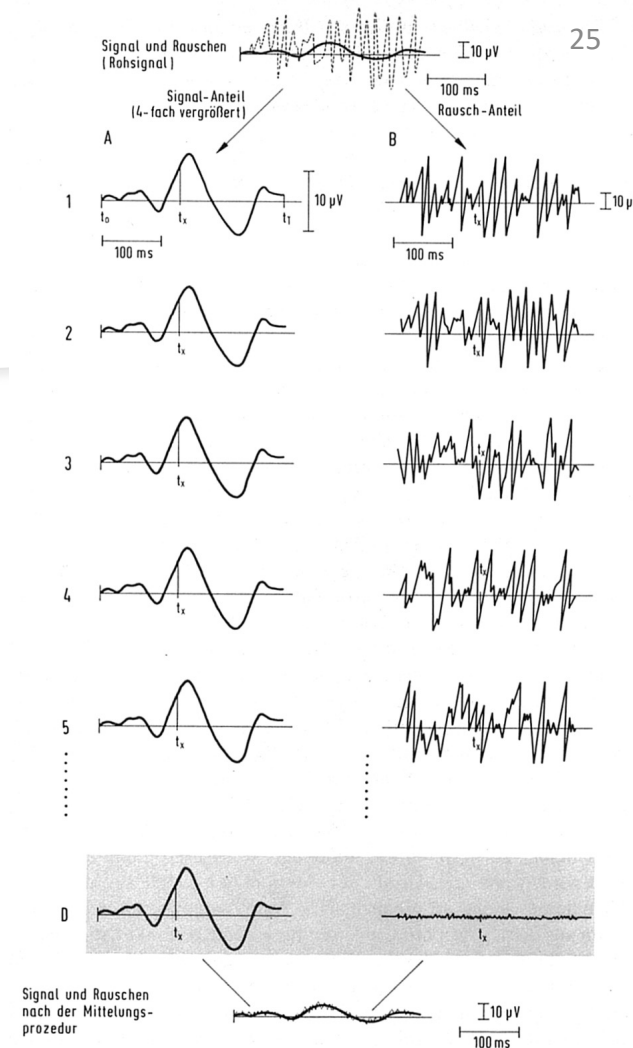
To solve these low-amplitude potentials against the background of ongoing EEG and external noise, **signal averaging** is required.

EVOKED POTENTIAL (EP)

The signal is time-locked to the stimulus and most of the noise occurs randomly, allowing the noise to be averaged out with averaging of repeated responses.

Signal Averaging

- Not just one instance of the event of interest but many instances
- Epochs of brain activity are averaged together
- The "random" activity washes out during averaging, whereas the brain activity of interest stays in the signal
- Through this signal-averaging procedure, it is possible to isolate the brain response that is specifically elicited in response to some event of interest



$$\hat{u}(t) = \frac{1}{N} \sum_{i=1}^N y_i(t)$$

Types of EP

According to the type of stimulation different brain circuits are activated.

The most common evoked potentials are **sensory** and **motor** evoked potentials. The sensory evoked potentials depending on the kind of the external stimulus are categorized as: *visual, auditory and somatosensory*.

- EP procedure
- EP components named by their polarity and peak latency (in ms)